Methods Practice SAC3

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CAS + No CAS | 10m Reading | 1h 40m Writing | 58 marks

*All questions are No CAS except for those labelled with '***CAS ACTIVE***'.*
Please only use the CAS for these questions.

Question 1 (7 marks)
A student takes a multiple choice exam in which each question has 5 options, exactly one of which is correct.

If the student knows the correct answer, then they select that option. Otherwise, they just randomly select one of the 5 options.

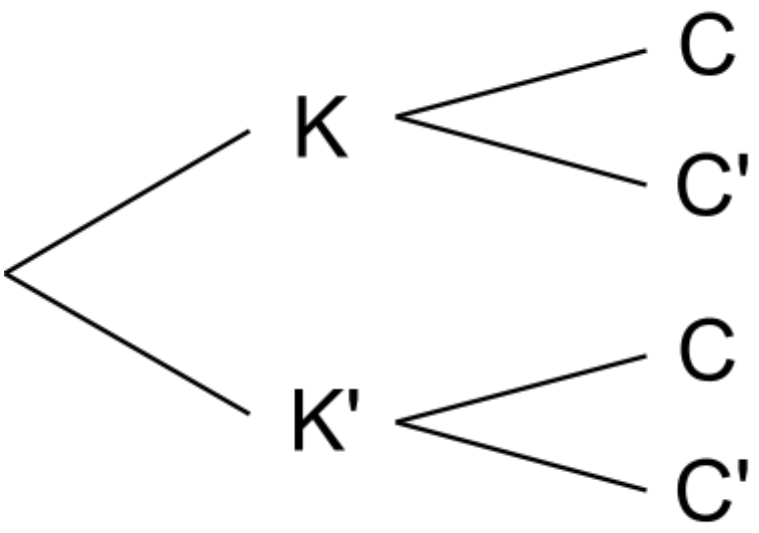
Suppose the student knows the answer to 50% of the questions.

For any given question, we define the events C and K as:
C: the student picked the **c**orrect option.
K: the student **k**new the answer.

a. Determine each of: (2 marks)

$Pr(K)$ $Pr(C \mid K)$ $Pr(C \mid K')$

b. Model these probabilities, and their complements $Pr(K')$, $Pr(C' \mid K)$ and $Pr(C' \mid K')$, on the tree diagram below: (1 mark)



c. Hence, or otherwise, find the probability that the student picked the correct option. That is, find $Pr(C)$. (2 marks)

d. Given the student picked the correct option, what is the probability that they actually knew the answer? (2 marks)

Question 2 (7 marks)

The events A and B satisfy the following equations:

$$Pr(A) = \frac{1}{3} \qquad Pr(A \cap B) = \frac{1}{4} \qquad Pr(A' \cap B) = \frac{1}{4}$$

With the help of Venn diagrams, or otherwise, find:

a. $Pr(A')$ (1 mark)

b. $Pr(B \mid A)$ (2 marks)

c. $Pr(B)$ (2 marks)

d. $Pr(A' \mid B')$ (2 marks)

Question 3 (8 marks)

Neap 2020 Exam 1 Q6

A phone dealer has eight new J-phones in stock. It is found that three of these phones have faulty batteries.

a. Two of the eight J-phones are selected at random.

i. Find the probability that at least one of the selected phones has a faulty battery. 2 marks

ii. Given that at least one of the selected phones has a faulty battery, find the probability that exactly one phone has a faulty battery. 2 marks

b. It is known that, overall, 20% of J-phones will have a faulty battery. The probability of any given phone having a faulty battery is independent to any other phone. A phone dealer receives a delivery of four new J-phones.

Find the probability that at least two of these phones will have faulty batteries. 2 marks

c. Working batteries have a lifespan, measured in minutes, after which they require recharging.

The lifespan of working batteries is a normally distributed variable, X, with a mean of 540 minutes and a standard deviation of 70 minutes. Let Z be the standard normal variable such that $Z \sim N(0, 1)$.

i. Find b such that $\Pr(X < b) = \Pr(Z > -1.5)$. 1 mark

ii. Correct to two decimal places, $\Pr(Z < -1) = 0.16$.

Find the probability that a randomly selected phone battery has a lifespan of greater than 470 minutes, given that it has a lifespan of less than 540 minutes. 1 mark

Question 4 (2 marks)

Neap 2021 Exam 1 Q6

An animal rescue shelter has a population of rabbits of different breeds. European rabbits form $\frac{3}{5}$ of the population. If a sample of size n is taken from the population, $\hat{P}$ represents the proportion of European rabbits in the sample.

Find the value of n such that the standard deviation of $\hat{P}$ is equal to $\frac{1}{10}$.

Question 5 (3 marks)

Neap 2021 Exam 1 Q5

The results of four independent trials show that $4\Pr(X=1)=\Pr(X=3)$, where the random variable X represents the number of successes. X is binomially distributed.

If p is the probability of success on any trial, find the value(s) of p.

Question 6 (7 marks)

Neap 2019 Exam 1 Q6

Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} kx^2(1-x) & 0 \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

a. Show that $k = 12$. 2 marks

b. Find the mean of X. 2 marks

c. Let Y be a continuous random variable with the probability density function

$$g(x) = \begin{cases} qx^2(p-x) & 0 \le x \le p \\ 0 & \text{elsewhere} \end{cases}$$

Given that $p, q > 0$ and the mean of Y is equal to $\frac{3}{10}$, find the value of p and q. 3 marks

Question 7 (11 marks) **CAS ACTIVE**

Neap 2019 Exam 2 Q2

The number of hamburgers sold at a fast-food restaurant on any given day is normally distributed with a mean of 120 and a standard deviation of 15.

a. Find the probability that on any given day more than 150 hamburgers are sold. Give your answer correct to four decimal places. 1 mark

b. If the restaurant sells at least x hamburgers on at least 75% of days, find the value of x to the nearest whole number. 1 mark

It is known that 70% of customers select a hamburger with cheese. A random sample of 100 customers was taken.

c. **i.** What is the probability, correct to four decimal places, that more than 75% of customers selected a hamburger with cheese? Do **not** use a normal approximation. 2 marks

ii. What is the answer to **part c.i.** if a normal approximation is used? Give your answer correct to four decimal places. 2 marks

In order to estimate the proportion of customers who like lettuce in their hamburger, another random sample was taken.

It found that out of 500 customers, 200 of them like lettuce in their hamburger.

d. Find a 95% confidence interval for the proportion of customers that like lettuce in their hamburger, correct to 4 decimal places. (1 mark)

e. Determine whether a 99% confidence interval would be narrower or wider than a 95% confidence interval. Explain why. (2 marks)

Another sample of n customers found a sample proportion of those who like lettuce in their hamburger of $\widehat{p} = 0.45$.

A 90% confidence interval was then constructed, accurate to 4 decimal places:

$$(0.4211, 0.4789)$$

f. Find the sample size n that resulted in this confidence interval.
Note: round n to the nearest integer. (2 marks)

Question 8 (13 marks) CAS ACTIVE

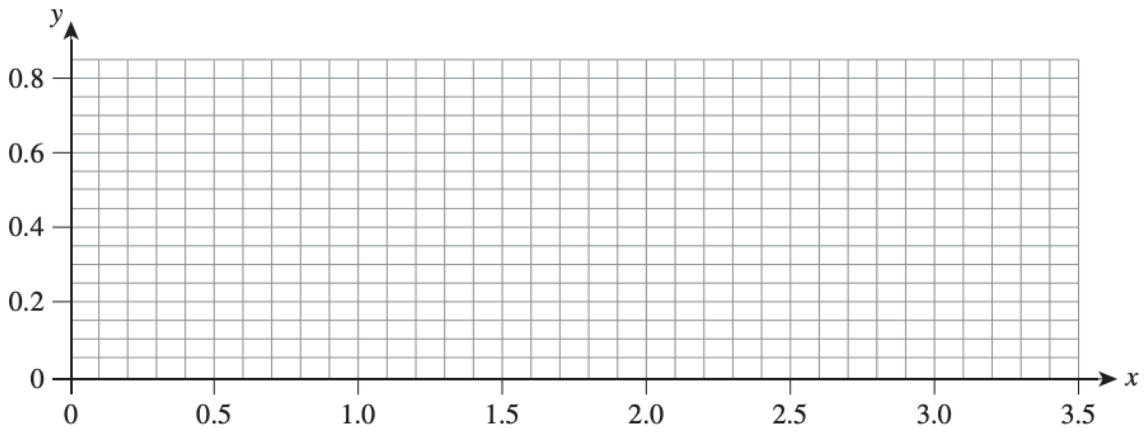
Neap 2019 Exam 2 Q4

Max is learning to juggle. The time that he can juggle before dropping a ball, t, in minutes, can be modelled by the probability density function $f(t)$.

$$f(t) = \begin{cases} t & 0 \le t \le \frac{2}{3} \\ k(t-3) & \frac{2}{3} < t \le 3 \\ 0 & \text{elsewhere} \end{cases}$$

a. Show that $k = -\frac{2}{7}$. 3 marks

b. Sketch the graph of f on the axes provided below. 2 marks



c. Find the probability that, given Max juggles for longer than 30 seconds, he juggles for at least 2 minutes. 3 marks

d. Find a such that $\Pr(T > a) = 0.6$. 2 marks

Max deems an individual attempt where he juggles for at least two minutes as "successful'.

Assume that the amount of time that Max juggles in any given attempt is independent of any other attempt.

e. Find the probability that any given attempt is ‘successful’. (1 mark)

f. Using $invBinomN$, or otherwise, find the minimum number of attempts required for Max to ensure that he has a greater than 70% chance of **at least three** ‘successful’ attempts. (2 marks)